

NAME: CHRISTIANA OFORI ANSONG

COHORT: SEPTEMBER 2026

COURSE: GRC

ASSIGNMENT : INTRODUCTION TO NETWORKING

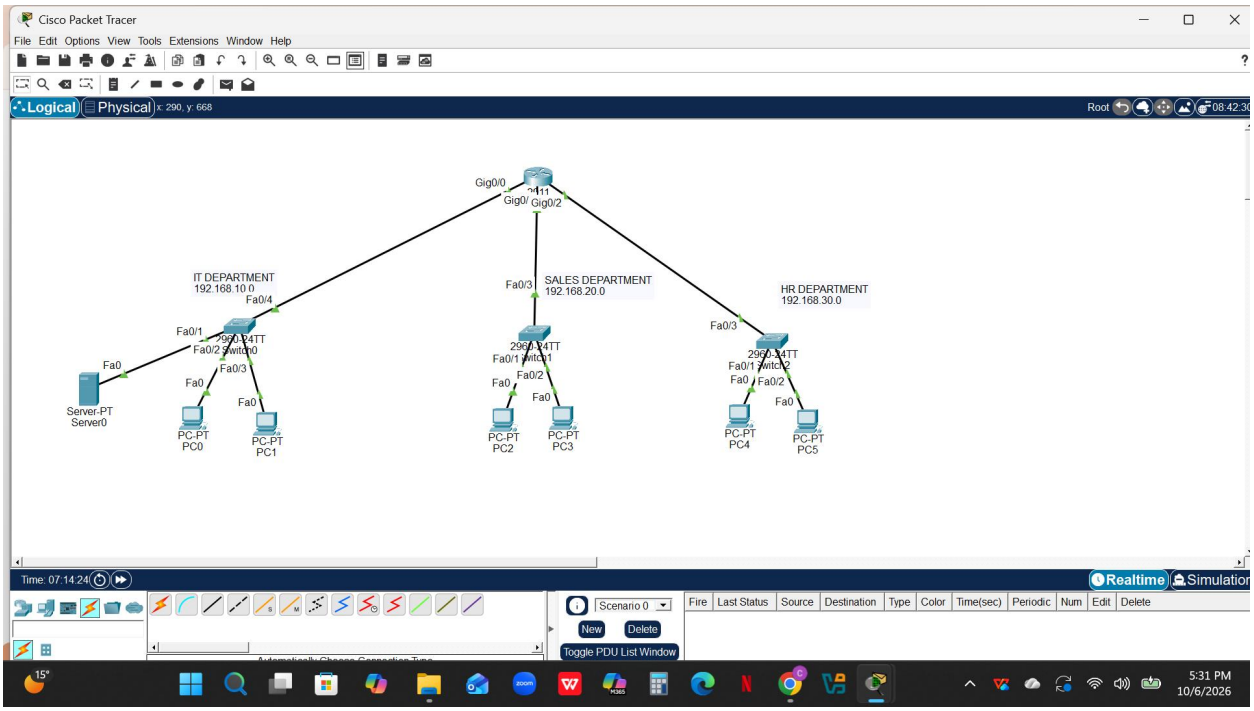
DATE: 8TH OCTOBER, 2026

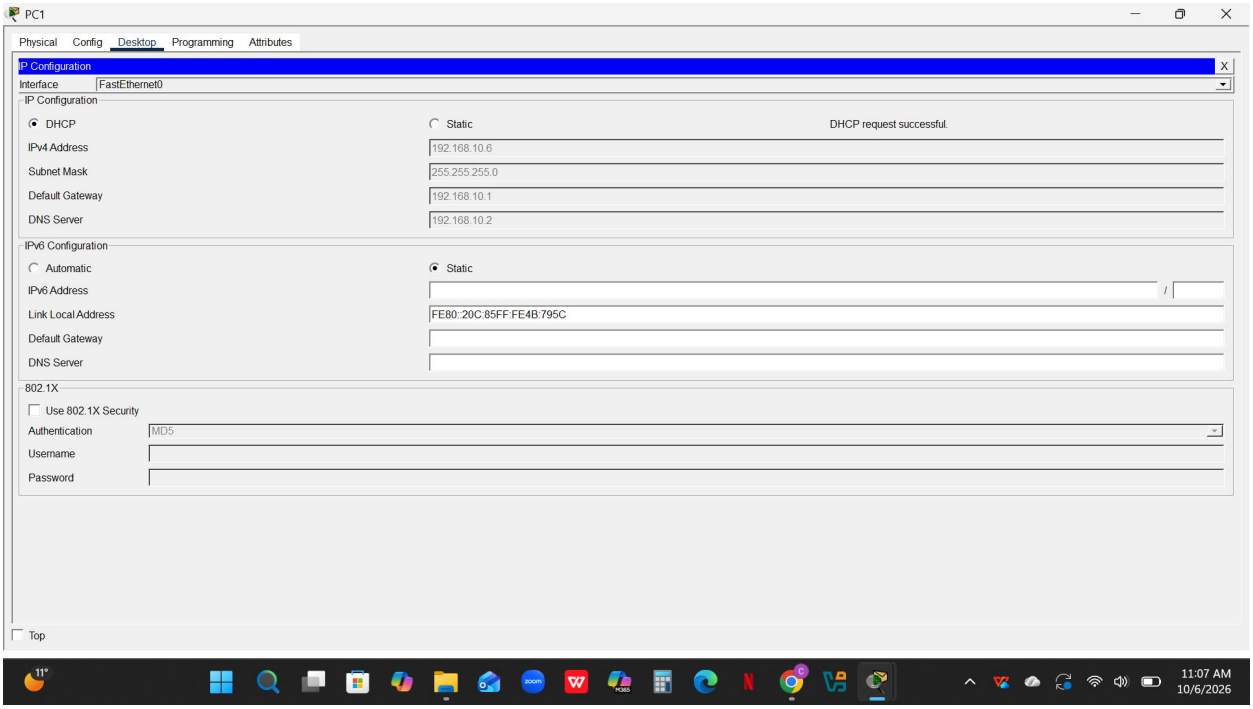
Business Problem – NetCore Systems Ltd.

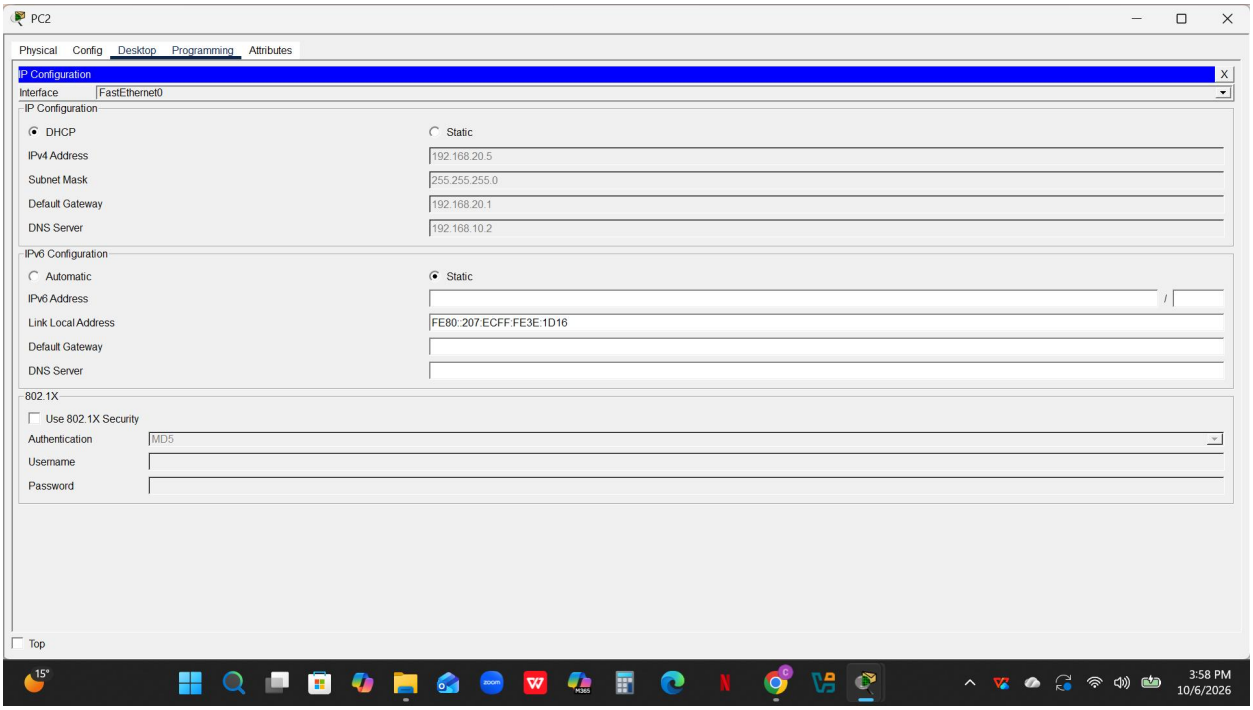
As a Network Technician, I identified that NetCore Systems Ltd. currently lacks a structured and secure LAN connecting its IT, Sales, and HR departments. This results in inefficient communication, slower operations, limited network management, and increased security risks.

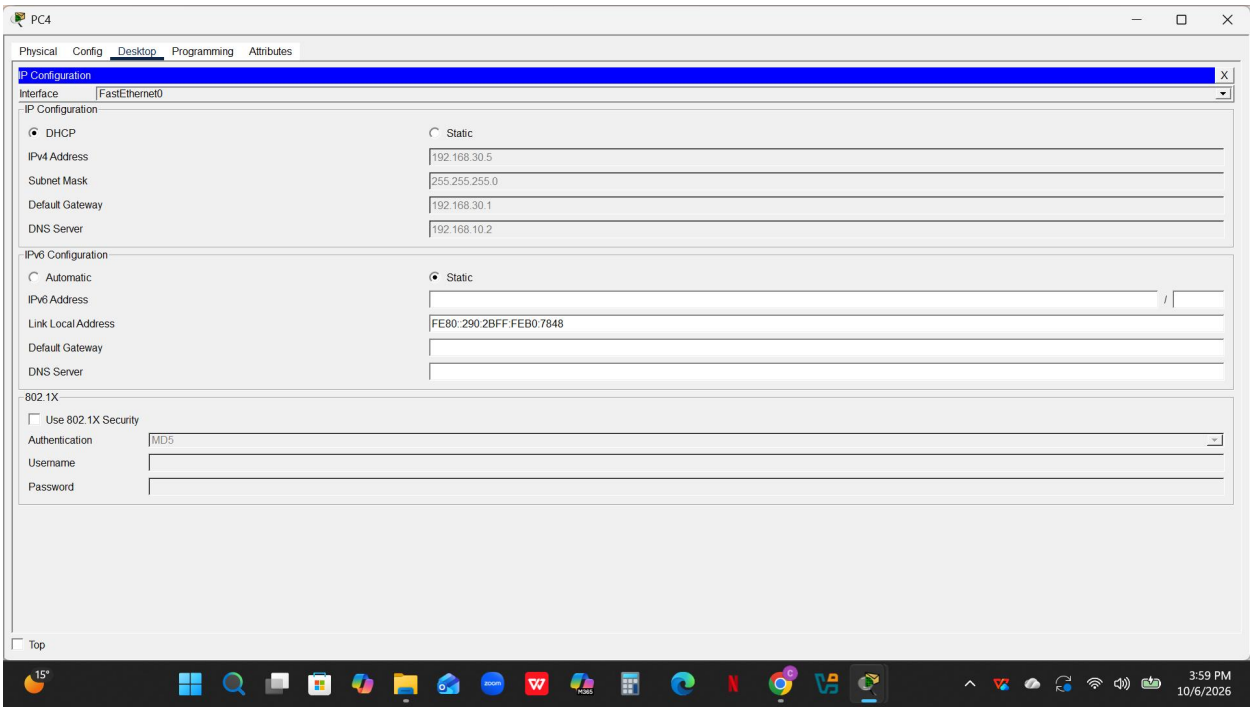
The company also lacks centralized IP addressing, DHCP, and network services such as FTP and HTTP, making the network difficult to manage and troubleshoot. In addition, there are no effective access controls to protect sensitive departmental resources.

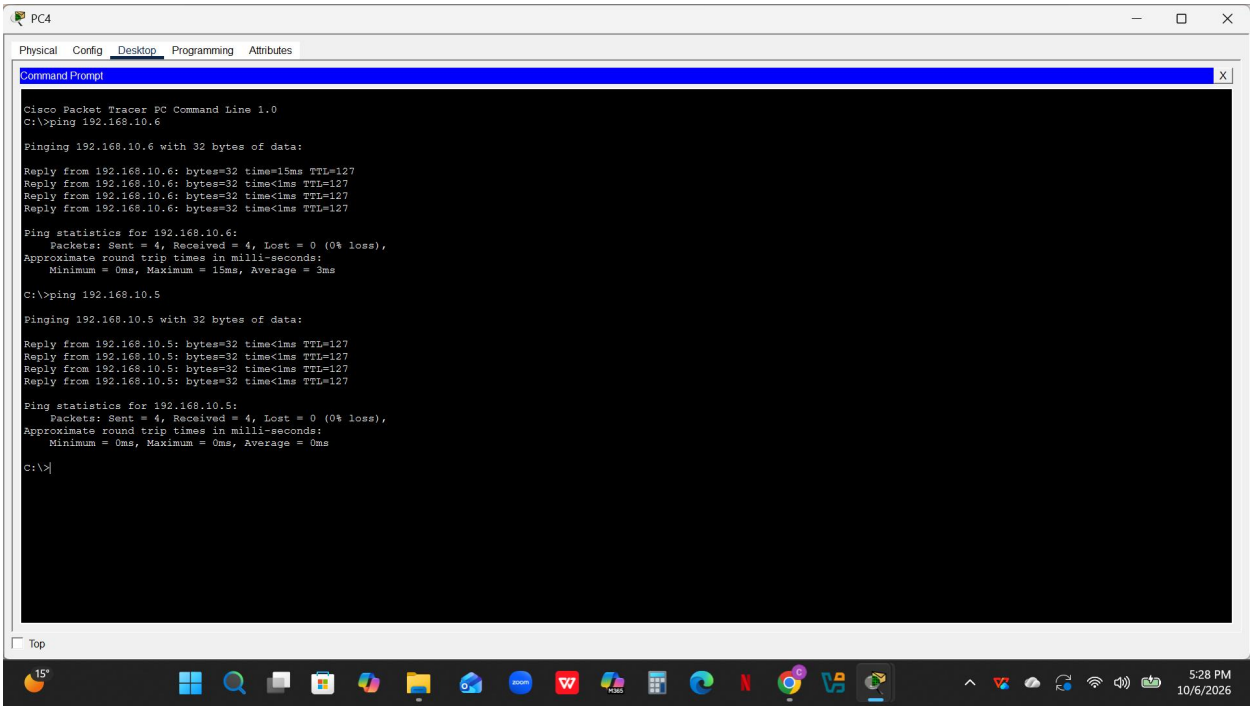
The proposed network solution will establish a structured Cisco Packet Tracer LAN with appropriate IP addressing, centralized services, inter-department communication, and ACL-based security controls. This will improve collaboration, simplify network management, strengthen security, and provide a scalable foundation for NetCore Systems Ltd.'s future growth.




PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.10.6
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
192.168.10.2
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20C:85FF:FE4B:795C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
11°
11:07 AM
10/6/2026


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.5
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
192.168.10.2
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::207:ECFF:FE3E:1D16
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
15°
3:58 PM
10/6/2026


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.30.5
Subnet Mask
255.255.255.0
Default Gateway
192.168.30.1
DNS Server
192.168.10.2
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::290:2BFF:FEB0:7848
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
3:59 PM
10/6/2026


PC4
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.6
Pinging 192.168.10.6 with 32 bytes of data:
Reply from 192.168.10.6: bytes=32 time=15ms TTL=127
Reply from 192.168.10.6: bytes=32 time<1ms TTL=127
Reply from 192.168.10.6: bytes=32 time<1ms TTL=127
Reply from 192.168.10.6: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.10.6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 15ms, Average = 3ms
C:\>ping 192.168.10.5
Pinging 192.168.10.5 with 32 bytes of data:
Reply from 192.168.10.5: bytes=32 time<1ms TTL=127
Reply from 192.168.10.5: bytes=32 time<1ms TTL=127
Reply from 192.168.10.5: bytes=32 time<1ms TTL=127
Reply from 192.168.10.5: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.10.5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
5:28 PM
10/6/2026

```
C:\>ping 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

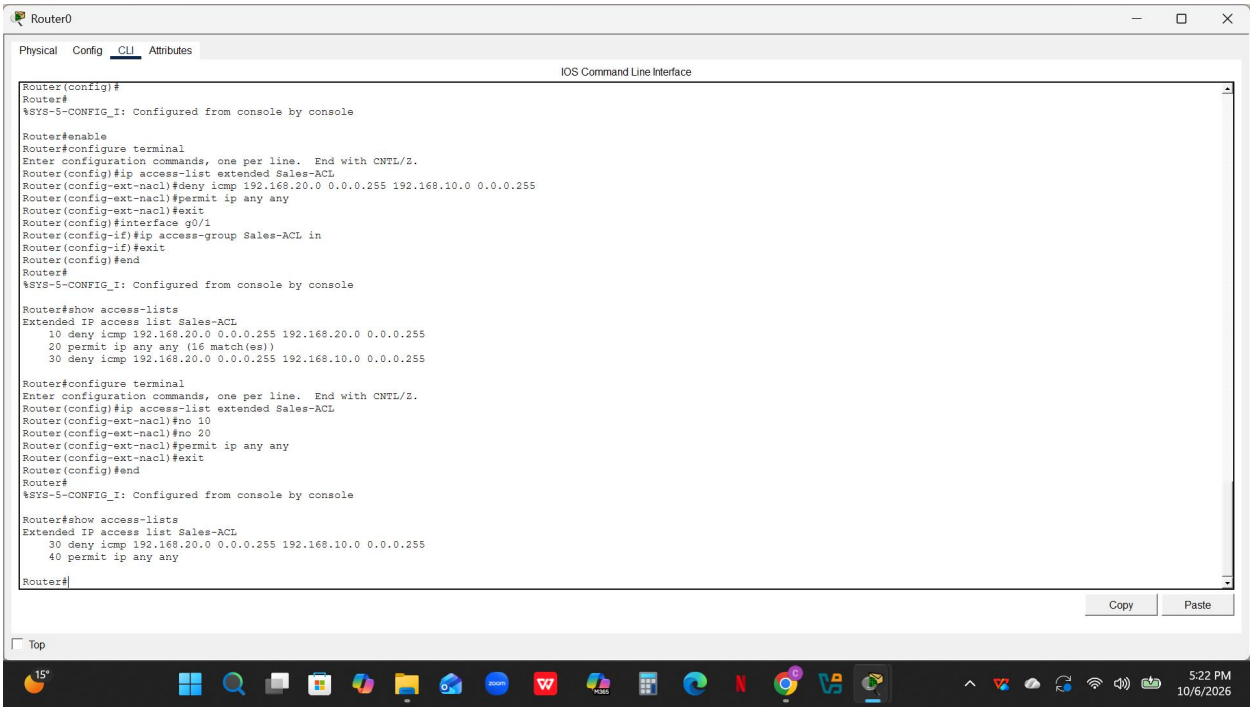
Ping statistics for 192.168.10.6:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

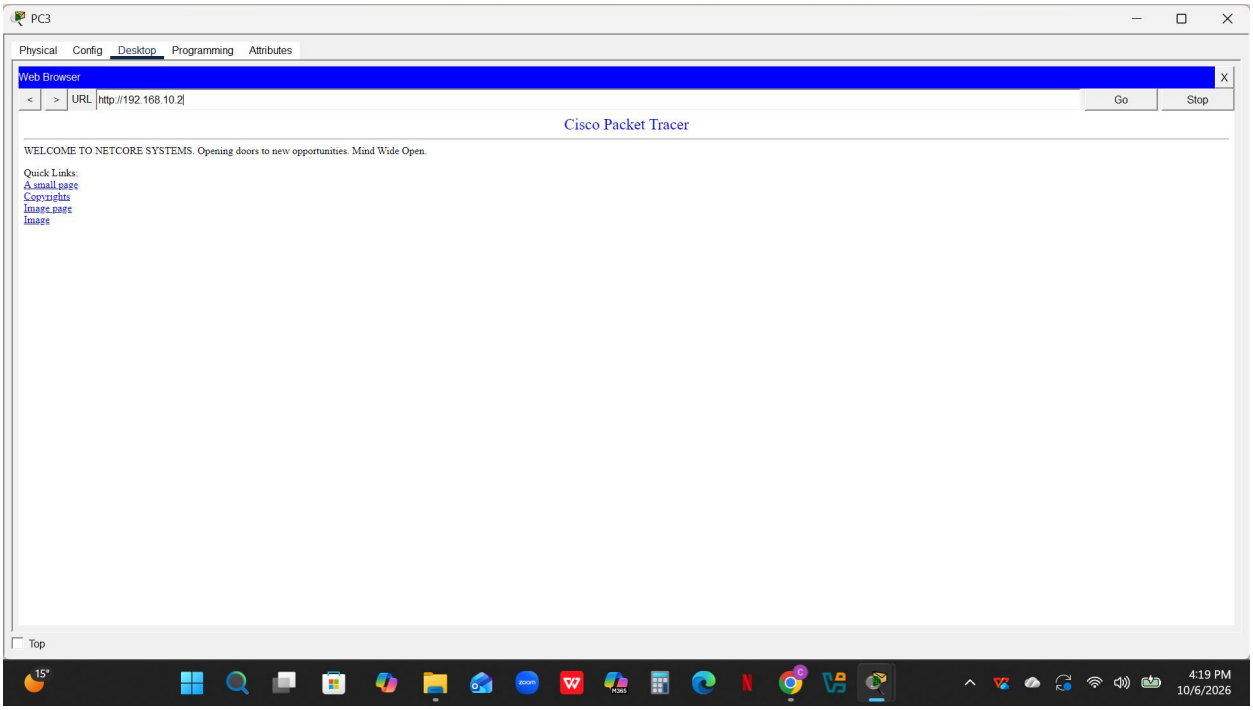
C:\>ping 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.6:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```





SUBMISSIONS

1st Submission: The image shows the Cisco Packet Tracer workspace with a network topology consisting of three departments (IT Dept, Sales Dept, and the HR Dept) connected through a central router named Netcore-Router which acts as the main gateway routing traffic between three (3) distinct departments. The IT department utilizes the 192.168.10.0 network, comprising two PCs and a server, whereas the Sales and HR departments use the 192.168.20.0 and 192.168.30.0 networks, respectively. These networks connect to a central Cisco 2911 router, which manages traffic between them. This design improves network organization, makes administration easier, and provides a foundation for implementing security policies and controlling communication between departments.All the link lights showing green indicates connections between the PCs, switches and the router are up and active and that devices can communicate .

2nd Submission: The images shows the IP configuration window for a specific end-device(PC) inside the Cisco Packet Tracer for each three department.This is the configuration interface used to assign network settings to a device network card. The PCs were configured to get their network settings automatically via DHCP, meaning, they successfully requested and received their networks identity from the DHCP server on the network

3rd Submission : This shows a successful ping command prompt window inside Cisco Packet Tracer. The command executed was ping 192.168.10.5 and 192.168.10.6. This means the HR Department testing its connection on both of the PCs in the IT Department subnet was a success. The computer sent 4 test Packets, The destination machines received the 4packets and successful sent them back and zero percent loss was recorded meaning no data packets dropped, no misconfigured subnets, or blocking firewall rules between these two departments (HR & IT Dept.)

4th Submission: This shows a failed connectivity test in the command prompt. Unlike the previous successful test,this one hits a roadblock. The command executed was ping 192.168.10.6, that is trying to reach a device in the IT Dept subnet, but the result is a 100% packet loss with the specific error “Destination host unreachable” and the error is being sent back from 192.168.20.1 that is the gateway for the Sales Dept. This tells us that, the test was done from one of the Sales PCs.

5th Submission: This configuration shows the creation of an Extended Access Control List (ACL) called Sales-ACL on the router. The ACL is designed to control traffic between the Sales network, 192.168.20.0/24, and the IT network, 192.168.10.0/24. The deny rule blocks traffic from Sales to IT, while the permit ip any any rule allows all other traffic. This provides basic network segmentation and prevents unauthorized communication between the Sales and IT departments. In summary, Sales to IT blocked while Sales to other networks allowed and the purpose is to improve network security and departmental isolation. The key command used is : deny IP 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255.This specifically denies traffic from the Sales subnet to the IT subnet.

6th Submission: This screenshot demonstrates successful HTTP connectivity testing in Cisco Packet Tracer. PC3 accesses the web server using the IP address 192.168.10.2, and the NetCore Systems webpage loads successfully. This confirms that HTTP traffic is permitted and that the PC can communicate with the server across the network. This test shows that the network configuration and permitted traffic are functioning correctly.